FINS3616 International Business Finance Term

Term 2, 2026

Week 4 Reading Guide and Tutorial Questions

Chs 7 and 8: Currency Futures and Options and Interest Rate and Currency Swaps

Week 4 Readings (Shapiro)

Chapter 7 – *Currency Futures and Options Markets*

7.1 Futures Contracts

7.2 Currency Options

7.3 Reading Currency Futures and Options Prices

Chapter 8 –Interest Rate and Currency Swaps

What is a currency swap?

Tutorial Questions

Chapter 7:

Conceptual Questions 2, 3, 4, 5

Futures: Problems 1, 2, 3

Options: Problems 5, 6

Chapter 8

Questions 2 and 3

Chapter 8, Problem 3

Chapter 7

Chapter 7, Question 2

2. What are the basic differences between forward and futures contracts? Between futures and options contracts?

ANSWER. The basic differences between forward and futures contracts are described in Section 3.1. The most important difference between these two contracts and an options contract is that a buyer of a forward or futures contract must take delivery, while the buyer of an options contract has the right but not the obligation to complete the contract.

Chapter 7, Question 3

3. A forward market already existed, so why was it necessary to establish currency futures and currency options contracts?

ANSWER. A currency futures market arose because private individuals were unable to avail themselves of the forward market. Currency options are partly a response to individuals and firms who would like to eliminate some currency risk while at the same time preserving the possibility of earning a windfall profit from favourable movements in the exchange rate. Options also enable firms bidding on foreign projects to lock in the home currency value of their bid without exposing themselves to currency risk if their bid is rejected.

Chapter 7, Question 4

4. Suppose that Texas Instruments must pay a French supplier €10 million in 90 days.

a. Explain how TI can use currency futures to hedge its exchange risk. How many futures contracts will TI need to fully protect itself?

ANSWER. TI can hedge its exchange risk by buying euro futures contracts whose expiration date is the closest to the date on which it must pay its French supplier. Given a contract size of €125,000, TI must buy 10,000,000/125,000 = 80 futures contracts to hedge its euro payable.

b. Explain how TI can use currency options to hedge its exchange risk. How many options contracts will TI need to fully protect itself?

ANSWER. TI can hedge its exchange risk by buying euro call options contracts whose expiration date is the closest to the date on which it must pay its French supplier. Given a contract size of €62,500, TI must buy 10,000,000/62,500 = 160 options contracts to hedge its payable.

c. Discuss the advantages and disadvantages of using currency futures versus currency options to hedge TI's exchange risk.

ANSWER. A futures contract is most valuable when the quantity of foreign currency being hedged is known, as in the case here. An option contract is most valuable when the quantity of foreign currency is unknown. Other things being equal, therefore, TI should use futures contracts to hedge its currency risk. However, TI must honor its futures contracts even if the spot rate at settlement is less than the futures price. In contrast, TI can choose not to exercise currency call options if the call price exceeds the spot price. Although this feature is an advantage of currency options, it is fully priced out in the market via the call premium. Hence, options are not unambiguously better than futures. In this case, since the quantity of the future French franc outflow is known, TI should use currency futures to hedge its risk.

Chapter 7, Question 5

5. Suppose that Bechtel Group wants to hedge a bid on a Japanese construction project. But because the yen exposure is contingent on acceptance of its bid, Bechtel decides to buy a put option for the ¥15 billion bid amount rather than sell it forward. In order to reduce its hedging cost, however, Bechtel simultaneously sells a call option for ¥15 billion with the same strike price. Bechtel reasons *that it wants to protect its downside risk on the contract and is willing to sacrifice the upside potential in order to collect the call premium. Comment on Bechtel's hedging strategy.*

ANSWER. The combination of buying a put option and selling a call option at the same strike price is equivalent to selling ¥15 billion forward at a forward rate equal to the strike price on the put and call options. That is, Bechtel is no longer holding an option; it is now holding a forward contract. If the yen appreciates and Bechtel loses its bid, it will face an exchange loss equal to 15 billion x (actual spot rate - exercise price).

Chapter 7, Problem 1

1. On Monday morning, an investor takes a long position in a pound futures contract that matures on Wednesday afternoon. The agreed upon price is $1.78/£ for £62,500. At the close of trading on Monday, the futures price has risen to $1.79/£. At Tuesday close, the price rises further to $1.80/£. At Wednesday close, the price falls to $1.785/£, and the contract matures. The investor takes delivery of the pounds at the prevailing price of $1.785/£. Detail the daily settlement process (This Table is taken from Chapter 7 of the Textbook). What will be the investor's profit (loss)?

ANSWER

Time	Action	Cash Flow
Monday morning	Investor buys pound futures contract that matures in two days. Price is $1.78.	None
Monday close	Futures price rises to $1.79. Contract is marked-to-market.	Investor receives 62,500 x (1.79 - 1.78) = $625.
Tuesday close	Futures price rises to $1.80. Contract is marked-to-market.	Investor receives 62,500 x (1.80 - 1.79) = $625.
Wednesday close	Futures price falls to $1.785. (1) Contract is marked-to-market. (2) Investor takes delivery of £62,500.	(1) Investor pays 62,500 x (1.80 -1.785) = $937.50 (2) Investor pays 62,500 x 1.785 = $111,562.50.

Net profit is $1,250 - 937.50 = $312.50.

Chapter 7, Problem 2

2. Suppose that the forward ask price for March 20 on euros is $0.9127 at the same time that the price of IMM euro futures for delivery on March 20 is $0.9145. How could an arbitrageur profit from this situation? What will be the arbitrageur's profit per futures contract (size of contract is €125,000)?

ANSWER. Since the futures price exceeds the forward rate, the arbitrageur should sell futures contracts at $0.9145 and buy euro forward in the same amount at $0.9127. The arbitrageur will earn 125,000 (0.9145 - 0.9127) = $225 per euro futures contract arbitraged.

Chapter 7, Problem 3

3. Suppose that Dell buys a Swiss franc futures contract (contract size is SFr 125,000) at a price of \$0.83. If the spot rate for the Swiss franc at the date of settlement is SFr 1 = \$0.8250, what is Dell's gain or loss on this contract?

ANSWER. Dell has bought Swiss francs worth \$0.8250 at a price of \$0.83. Thus, it has lost \$0.005 per franc for a total loss of 125,000 x .005 = \$625.

Chapter 7, Problem 5

5. Citigroup sells a call option on euros (contract size is €500,000) at a premium of \$0.04 per euro. If the exercise price is \$0.91/€ and the spot price of the euro at the date of expiration is \$0.93/€, what is Citigroup's profit (loss) on the call option?

ANSWER. Since the spot price of \$0.93 exceeds the exercise price of \$0.91, Citigroup's counterparty will exercise its call option, causing Citigroup to lose 2¢ per euro. Adding in the 4¢ call premium it received gives Citigroup a net profit of 2¢ per euro on the call option for a total gain of .02 x 500,000 = \$10,000.

Chapter 7, Problem 6

6. Suppose you buy three June PHLX call options with a 90 strike price at a price of 2.3 (¢/€). (Contract size is €62,500).

a. *What would be your total dollar cost for these calls, ignoring broker fees?*

ANSWER. With each call option being for €62,500, the three contracts combined are for €187,500. At a price of 2.3¢/€, the total cost is therefore 187,500 x \$0.023 = \$4,312.50.

b. *After holding these calls for 60 days, you sell them for 3.8 (¢/€). What is your net profit on the contracts assuming that brokerage fees on both entry and exit were \$5 per contract and that your opportunity cost was 8% per annum on the money tied up in the premium?*

ANSWER. The net profit would be 1.5¢/€ (3.8 - 2.3) for a total profit before expenses of \$2,812.50 (0.015 x 187,500). Brokerage fees totalled \$10 per contract or \$30 overall. The opportunity cost would be \$4,312.50 x 0.08 x 60/365 = \$56.71. After deducting these expenses (which total \$86.71), the net profit is \$2,725.79.

Chapter 8

Chapter 8, Question 2

2. *What is a currency swap?*

ANSWER. A currency swap involves the exchange of principal plus interest payments in one currency for equivalent payments in another currency.

Chapter 8, Question 3

3. Comment on the following statement. "In order for one party to a swap to benefit, the other party must lose."

ANSWER. Given that both parties to the swap freely enter into the swap transaction, both must perceive benefits. The tax, financial market, and regulatory system arbitrage benefits associated with swaps are shared by both parties.

Chapter 8, Problem 3

2. In May 1988, Walt Disney Productions sold to Japanese investors a 20 year stream of projected yen royalties from Tokyo Disneyland. The present value of that stream of royalties, discounted at 6 percent (the return required by the Japanese investors), was ¥93 billion. Disney took the yen proceeds from the sale, converted them to dollars, and invested the dollars in bonds yielding 10 percent. According to Disney's chief financial officer, Gary Wilson, "In effect, we got money at a 6 percent discount rate, reinvested it at 10 percent, and hedged our royalty stream against yen fluctuations all in one transaction."

a. At the time of the sale, the exchange rate was ¥124 = $1. What dollar amount did Disney realize from the sale of its yen proceeds?

ANSWER. Disney realized 93,000,000,000/124 = $750,000,000 from the sale of its future yen proceeds.

b. Demonstrate the equivalence between Walt Disney's transaction and a currency swap.

ANSWER. In a currency/interest rate swap, one party trades a stream of payments in one currency, at one interest rate, for a stream of payments in a second currency, at a second interest rate. Disney's stream of yen royalties can be treated as a yen bond, which it traded for a dollar bond, with dollar payments. The only difference between the Disney swap and a traditional swap is that the latter usually involve cash outflows whereas the Disney swap involves cash inflows.

c. Comment on Gary Wilson's statement. Did Disney achieve the equivalent of a free lunch through its transaction?

ANSWER. Gary Wilson is committing the economist's unpardonable sin: He is comparing apples with oranges, in this case, a 6% yen interest rate with a 10% dollar interest rate. The international Fisher effect tells us that the most likely reason that the yen interest rate is 4 percentage points less than the equivalent dollar interest rate is because the market expects the dollar to depreciate by about 4% annually against the yen.